Chapter 13: Probability

Practice Set A: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** Board-pattern — 1 mark

If $P(A|B) = 0.3$, $P(A) = 0.4$ and $P(B) = 0.8$, then $P(B|A)$ is:

(A) 0.6 **(B)** 0.3

(C) 0.06 **(D)** 0.4

Formula / idea: Multiplication rule: $P(A \cap B) = P(B)\,P(A|B) = P(A)\,P(B|A)$.

1 Find $P(A \cap B)$ by the multiplication rule.

$$P(A \cap B) = P(B) \cdot P(A|B) = 0.8 \times 0.3 = 0.24$$

2 Now divide by $P(A)$. ✓ 1

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.24}{0.4} = 0.6$$

3 Pick the matching option.

Option (A)

Answer **(A)** 0.6

Watch out: Answering 0.3: $P(B|A)$ and $P(A|B)$ are not the same.

2 MCQ PYQ-style 1 mark

If E and F are two independent events such that $P(E) = \frac{2}{3}$ and $P(F) = \frac{3}{7}$, then $P(E|\bar{F})$ is equal to:

(A) $\frac{1}{6}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) $\frac{7}{9}$

Formula / idea: If E and F are independent, then E and $\bar{F}$ are also independent.

1 E, F independent, so $E, \bar{F}$ are independent.
$P(E \cap \bar{F}) = P(E)\,P(\bar{F})$

2 Apply the conditional probability formula. ✓ 1
$P(E|\bar{F}) = \frac{P(E)\,P(\bar{F})}{P(\bar{F})} = P(E) = \frac{2}{3}$

3 Pick the matching option.
Option (C)

Answer **(C)** $\frac{2}{3}$

Watch out: Answering $P(\bar{F}) = \frac{4}{7}$ or multiplying: for independent events the condition does not change $P(E)$.

3 MCQ CBSE SQP 2025-26 1 mark

A person observed the first 4 digits of your 6-digit PIN. What is the probability that the person can guess your PIN?

(A) $\frac{1}{81}$

(B) $\frac{1}{100}$

(C) $\frac{1}{90}$

(D) 1

Formula / idea: Given the first 4 digits, only the last 2 digits are unknown.

1 Each of the last 2 digits can be any of $0, 1, \ldots, 9$.
$\text{Possible endings} = 10 \times 10 = 100$

2 Only one ending is correct. ✓ 1

$$P(\text{correct guess}) = \frac{1}{100}$$

3 Pick the matching option.

Option (B)

Answer (B) $\frac{1}{100}$

Watch out: Choosing $\frac{1}{90}$ or $\frac{1}{81}$: PIN digits can repeat and can be 0.

4 MCQ CBSE SQP 2024-25 1 mark

For any two events A and B, if $P(\bar{A}) = \frac{1}{2}$, $P(\bar{B}) = \frac{2}{3}$ and $P(A \cap B) = \frac{1}{4}$, then $P(\bar{A}|\bar{B})$ equals:

(A) $\frac{3}{8}$

(B) $\frac{8}{9}$

(C) $\frac{5}{8}$

(D) $\frac{1}{4}$

Formula / idea: $P(\bar{A} \cap \bar{B}) = P\big((A \cup B)'\big) = 1 - P(A \cup B)$.

1 Find $P(A)$ and $P(B)$.

$$P(A) = 1 - \frac{1}{2} = \frac{1}{2}, \quad P(B) = 1 - \frac{2}{3} = \frac{1}{3}$$

2 Addition rule.

$$P(A \cup B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$$

3 Use De Morgan's law.

$$P(\bar{A} \cap \bar{B}) = 1 - \frac{7}{12} = \frac{5}{12}$$

4 Apply the conditional probability formula. ✓ 1

$$P(\bar{A}|\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{5/12}{2/3} = \frac{5}{8}$$

5 Pick the matching option.

Option (C)

Answer (C) $\frac{5}{8}$

Watch out: Writing $P(\bar{A} \cap \bar{B}) = 1 - P(A \cap B)$: it is $1 - P(A \cup B)$.

5 MCQ Board-pattern 1 mark

A random variable X has the probability distribution X: $-1,\ 0,\ 1,\ 2$ with $P(X)$: $0.2,\ k,\ 2k,\ 0.2$. The value of k is:

(A) 0.1 **(B)** 0.2

(C) 0.3 **(D)** 0.6

Formula / idea: In a probability distribution, $\sum P(X = x_i) = 1$.

1 Sum of all probabilities is 1.
$0.2 + k + 2k + 0.2 = 1$

2 Solve for k.
$3k = 0.6 \Rightarrow k = 0.2$ ✓ 1

3 Pick the matching option.
Option (B)

Answer **(B)** 0.2

Watch out: Stopping at $3k = 0.6$ and choosing 0.6: divide by 3.

6 Assertion-Reason Board-pattern 1 mark

Assertion (A): Events A and B with $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.8$ are independent.
Reason (R): Two events A and B are independent if and only if $P(A \cap B) = P(A)\,P(B)$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) **Assertion (A) is false, but Reason (R) is true.**

Formula / idea: A, B independent $\iff P(A \cap B) = P(A)\,P(B)$.

1 Find $P(A \cap B)$ from the addition rule.
$P(A \cap B) = 0.5 + 0.4 - 0.8 = 0.1$

2 Compare with $P(A)\,P(B)$.
$P(A)\,P(B) = 0.5 \times 0.4 = 0.2 \neq 0.1$

3 $P(A \cap B) \neq P(A)\,P(B)$, so **Assertion (A) is false**.

4 Reason is the definition of independent events, so **Reason (R) is true**.

5 A is false, so the question of R explaining A does not arise.

6	Pick the matching option. Option (D)	✓ 1

Answer **(D) A is false, but R is true**

Watch out: Taking $P(A \cap B) = 0.8$: that is $P(A \cup B)$, not the intersection.

7 **Short answer** Board-pattern 2 marks

A box of oranges is inspected by examining three randomly selected oranges drawn without replacement. If all the three oranges are good, the box is approved for sale, otherwise it is rejected. Find the probability that a box containing 15 oranges, out of which 12 are good and 3 are bad, will be approved for sale.

Formula / idea: Multiplication theorem: $P(E_1 \cap E_2 \cap E_3) = P(E_1)\,P(E_2|E_1)\,P(E_3|E_1 \cap E_2)$.

1	Let E_1, E_2, E_3: first, second, third orange is good.	
2	Write the multiplication theorem. $P(E_1 \cap E_2 \cap E_3) = P(E_1)\,P(E_2|E_1)\,P(E_3|E_1 \cap E_2)$	✓ ½
3	12 good oranges in 15. $P(E_1) = \frac{12}{15}$	✓ ½
4	Without replacement: 11 good in 14, then 10 good in 13. $P(E_2|E_1) = \frac{11}{14}, \quad P(E_3|E_1 \cap E_2) = \frac{10}{13}$	✓ ½
5	Multiply and simplify. $P = \frac{12}{15} \times \frac{11}{14} \times \frac{10}{13} = \frac{1320}{2730} = \frac{44}{91}$	✓ ½
6	Probability that the box is approved $= \frac{44}{91}$.	

Answer $\frac{44}{91}$

Watch out: Using $\left(\frac{12}{15}\right)^3$: without replacement, the counts drop after each draw.

Level 2

18 marks

8 **Short answer** PYQ-style 3 marks

A bank offers loans to its customers on three types of interest: fixed rate, floating rate and variable rate. From past data, a customer avails a loan on fixed rate, floating rate and variable rate with probabilities 10%, 20% and 70% respectively. The probability that a customer defaults on repayment is 5%, 3% and 1% for fixed, floating and variable rate loans respectively. A customer is found to have defaulted on loan repayment. Find the probability that he availed the loan at a variable rate of interest.

Formula / idea: Bayes' theorem: $P(E_i|A) = \dfrac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$.

1 Let E_1, E_2, E_3: loan on fixed, floating, variable rate; A: customer defaults.

2 Write the prior probabilities. ✓ ½

$$P(E_1) = \frac{10}{100}, \quad P(E_2) = \frac{20}{100}, \quad P(E_3) = \frac{70}{100}$$

3 Write the probabilities of default. ✓ ½

$$P(A|E_1) = \frac{5}{100}, \quad P(A|E_2) = \frac{3}{100}, \quad P(A|E_3) = \frac{1}{100}$$

4 Find $P(A)$ by the theorem of total probability. ✓ 1

$$P(A) = \frac{10\times 5 + 20\times 3 + 70\times 1}{10000} = \frac{180}{10000} = \frac{18}{1000}$$

5 Apply Bayes' theorem for E_3. ✓ ½

$$P(E_3|A) = \frac{P(E_3)\,P(A|E_3)}{P(A)} = \frac{70/10000}{180/10000}$$

6 Simplify. ✓ ½

$$P(E_3|A) = \frac{70}{180} = \frac{7}{18}$$

7 Required probability $= \dfrac{7}{18}$.

Answer $P(E_3|A) = \dfrac{7}{18}$

Watch out: Answering $P(A|E_3) = 1\%$: the question asks for $P(E_3|A)$.

9 **Short answer** Board-pattern 2 marks

A fair die is thrown twice. Find the probability distribution of the number of sixes. Also find its mean.

Formula / idea: Mean $E(X) = \sum x_i\, p_i$.

1 Let X = number of sixes, so X takes values $0, 1, 2$. ✓ ½

$$P(\text{six}) = \frac{1}{6}, \quad P(\text{not six}) = \frac{5}{6}$$

2 Find each probability (throws are independent).
$P(X=0)=\frac{5}{6}\cdot\frac{5}{6}=\frac{25}{36},\ P(X=1)=2\cdot\frac{1}{6}\cdot\frac{5}{6}=\frac{10}{36},\ P(X=2)=\frac{1}{36}$

3 Write the probability distribution. ✓ 1

X	0	1	2
$P(X)$	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$

4 Find the mean. ✓ ½
$E(X)=0\cdot\frac{25}{36}+1\cdot\frac{10}{36}+2\cdot\frac{1}{36}=\frac{12}{36}=\frac{1}{3}$

5 Mean number of sixes $=\frac{1}{3}$.

Answer X**:** $0,1,2$ **with** $P(X)$**:** $\frac{25}{36},\frac{10}{36},\frac{1}{36}$**; mean** $=\frac{1}{3}$

Watch out: Taking $P(X=1)=\frac{5}{36}$: a six can come on the first or the second throw.

10 **Short answer** CBSE SQP 2024-25 3 marks

The probability that it rains today is 0.4. If it rains today, the probability that it will rain tomorrow is 0.8. If it does not rain today, the probability that it will rain tomorrow is 0.7. Let P_1 denote the probability that it does not rain today, P_2 the probability that it will not rain tomorrow if it rains today, P_3 the probability that it will rain tomorrow if it does not rain today, and P_4 the probability that it will not rain tomorrow if it does not rain today.

(i) Find the value of $P_1\times P_4-P_2\times P_3$. 2 marks

(ii) Calculate the probability of raining tomorrow. 1 mark

Formula / idea: $P(\bar{A}|E)=1-P(A|E)$; total probability: $P(A)=\sum P(E_i)P(A|E_i)$.

(i) 2 marks

1 Not raining today is the complement of raining today. ✓ ½
$P_1=1-0.4=0.6$

2 If it rains today, 'no rain tomorrow' is the complement. ✓ ½
$P_2=1-0.8=0.2$

3 If it does not rain today: given value and its complement. ✓ ½
$P_3=0.7,\quad P_4=1-0.7=0.3$

4 Substitute. ✓ ½
$P_1P_4-P_2P_3=0.6\times0.3-0.2\times0.7=0.18-0.14=0.04$

Answer (i) 0.04

(ii) 1 mark

1	Let E_1: rain today, E_2: no rain today, A: rain tomorrow. $P(E_1) = 0.4,\ P(E_2) = 0.6,\ P(A\|E_1) = 0.8,\ P(A\|E_2) = 0.7$	
2	Theorem of total probability. $P(A) = P(E_1)P(A\|E_1) + P(E_2)P(A\|E_2) = 0.4 \times 0.8 + 0.6 \times 0.7$	✓ ½
3	Simplify. $P(A) = 0.32 + 0.42 = 0.74$	✓ ½
4	Probability of rain tomorrow $= 0.74$.	

Answer (ii) 0.74

Watch out: Adding $0.8 + 0.7$ for (ii): each conditional must be weighted by $P(E_1)$ or $P(E_2)$.

11 **Short answer** CBSE SQP 2025-26 3 marks

Two students Mehul and Rashi are seeking admission in a college. The probability that Mehul is selected is 0.4 and the probability of selection of exactly one of them is 0.5. Chances of selection of them are independent of each other. Find the chances of selection of Rashi. Also find the probability of selection of at least one of them.

Formula / idea: For independent events, $P(M \cap R) = P(M)P(R)$, and M', R' are also independent.

1	Let M: Mehul selected, R: Rashi selected, $P(R) = x$. $P(M) = 0.4, \quad P(M') = 0.6, \quad P(R') = 1 - x$	✓ ½
2	Exactly one selected: M and not R, or not M and R. $P(M)P(R') + P(M')P(R) = 0.5$	✓ ½
3	Substitute and simplify. $0.4(1 - x) + 0.6x = 0.5 \ \Rightarrow\ 0.4 + 0.2x = 0.5$	✓ ½
4	Solve for x. $x = P(R) = 0.5$	✓ ½
5	At least one selected $= 1-$ neither selected. $P(\text{at least one}) = 1 - P(M')P(R') = 1 - 0.6 \times 0.5$	✓ ½
6	Simplify. $P(\text{at least one}) = 1 - 0.3 = 0.7$	✓ ½
7	Hence $P(\text{Rashi}) = 0.5$ and $P(\text{at least one}) = 0.7$.	

Answer $P(R) = 0.5$; $P(\text{at least one}) = 0.7$

Watch out: Writing 'exactly one' as $P(M) + P(R)$: that also counts the case when both are selected.

12 **Short answer** Board-pattern 2 marks

A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?

Formula / idea: $P(E|F) = \dfrac{P(E \cap F)}{P(F)}$.

1	Let F: sum is 6; list its outcomes. $F = \{(1,5),(2,4),(3,3),(4,2),(5,1)\}$	✓ ½
2	Let E: 4 appears at least once; find $E \cap F$. $E \cap F = \{(2,4),(4,2)\}$	✓ ½
3	Find the probabilities (36 outcomes in all). $P(F) = \dfrac{5}{36}, \quad P(E \cap F) = \dfrac{2}{36}$	
4	Apply the conditional probability formula. $P(E\|F) = \dfrac{2/36}{5/36} = \dfrac{2}{5}$	✓ 1
5	Required probability $= \dfrac{2}{5}$.	

Answer $\dfrac{2}{5}$

Watch out: Answering $\dfrac{11}{36}$: that is $P(E)$; the sum 6 is already given.

13 **Short answer** CBSE SQP 2024-25 3 marks

A random variable X can take all non-negative integral values and the probability that X takes the value r is proportional to 5^{-r}. Find $P(X < 3)$.

Formula / idea: $\sum P(X = r) = 1$; sum of an infinite G.P. $= \dfrac{a}{1-r}$ for $|r| < 1$.

1	Proportional to 5^{-r}, so put a constant k. $P(X = r) = \dfrac{k}{5^r}, \quad r = 0, 1, 2, \ldots$	✓ ½
2	Sum of all probabilities is 1. $k\left(1 + \dfrac{1}{5} + \dfrac{1}{5^2} + \ldots\right) = 1$	✓ ½
3	Sum the infinite G.P. with ratio $\dfrac{1}{5}$. $k \cdot \dfrac{1}{1 - \frac{1}{5}} = 1 \Rightarrow \dfrac{5k}{4} = 1 \Rightarrow k = \dfrac{4}{5}$	✓ ½

4	$X < 3$ means $X = 0, 1, 2$. $P(X < 3) = \frac{4}{5}\left(1 + \frac{1}{5} + \frac{1}{25}\right)$	✓ ½
5	Simplify. $P(X < 3) = \frac{4}{5} \cdot \frac{25 + 5 + 1}{25} = \frac{124}{125}$	✓ 1
6	Hence $P(X < 3) = \frac{124}{125}$.	

Answer $P(X < 3) = \frac{124}{125}$ **(with** $k = \frac{4}{5}$**)**

Watch out: Starting the values at $r = 1$: non-negative integers include $r = 0$.

14 **Short answer** Board-pattern 2 marks

A die is thrown once. Let E: "the number is even" and F: "the number is a prime number". Are E and F independent? Justify.

Formula / idea: E, F independent $\iff P(E \cap F) = P(E)\,P(F)$.

1	Write the sample space. $S = \{1, 2, 3, 4, 5, 6\}$	
2	Write E and find $P(E)$. $E = \{2, 4, 6\}, \quad P(E) = \frac{3}{6} = \frac{1}{2}$	✓ ½
3	Write F and find $P(F)$. $F = \{2, 3, 5\}, \quad P(F) = \frac{3}{6} = \frac{1}{2}$	✓ ½
4	Find $E \cap F$. $E \cap F = \{2\}, \quad P(E \cap F) = \frac{1}{6}$	✓ ½
5	Compare with $P(E)\,P(F)$. $P(E)\,P(F) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \neq \frac{1}{6}$	
6	Hence E and F are not independent.	✓ ½

Answer **No,** $P(E \cap F) = \frac{1}{6} \neq \frac{1}{4} = P(E)\,P(F)$**, so** E **and** F **are not independent**

Watch out: Counting 1 as a prime: the primes on a die are only 2, 3 and 5.

Level 3

25 marks

15 **Long answer** PYQ-style 5 marks

In an office three employees Jayant, Sonia and Oliver process incoming copies of a certain form. Jayant processes 50% of the forms, Sonia processes 20% and Oliver the remaining 30% of the forms. Jayant has an error rate of 0.06, Sonia has an error rate of 0.04 and Oliver has an error rate of 0.03. The manager selects a form at random from the day's output of processed forms. If the form selected has an error, find the probability that the form is not processed by Jayant.

Formula / idea: Bayes' theorem: $P(E_i|E) = \dfrac{P(E_i)P(E|E_i)}{\sum P(E_j)P(E|E_j)}$.

1	Let E_1, E_2, E_3: form processed by Jayant, Sonia, Oliver; E: form has an error.	✓ ½
2	Write the prior probabilities. $P(E_1) = 0.5, \quad P(E_2) = 0.2, \quad P(E_3) = 0.3$	✓ 1
3	Write the error rates. $P(E\|E_1) = 0.06, \quad P(E\|E_2) = 0.04, \quad P(E\|E_3) = 0.03$	✓ 1
4	Find $P(E)$ by the theorem of total probability. $P(E) = 0.5 \times 0.06 + 0.2 \times 0.04 + 0.3 \times 0.03$ $= 0.030 + 0.008 + 0.009 = 0.047$	✓ 1
5	Apply Bayes' theorem for Jayant. $P(E_1\|E) = \dfrac{P(E_1)P(E\|E_1)}{P(E)} = \dfrac{0.030}{0.047} = \dfrac{30}{47}$	✓ 1
6	'Not Jayant' is the complement. $P(\text{not Jayant}\|E) = 1 - \dfrac{30}{47} = \dfrac{17}{47}$	✓ ½
7	Required probability $= \dfrac{17}{47}$.	

Answer $P(\text{not Jayant}|E) = \dfrac{17}{47}$

Watch out: Answering $1 - 0.5 = 0.5$: the error is given, so Bayes' theorem must be used.

16 **Short answer** Board-pattern 3 marks

A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.

Formula / idea: Bayes' theorem with two cases: six occurs (E_1) or does not (E_2).

1	Let E_1: six occurs, E_2: six does not occur, A: he reports a six.	

2 Write the prior probabilities. ✓ 1

$$P(E_1) = \frac{1}{6}, \quad P(E_2) = \frac{5}{6}$$

3 He reports a six when truthful (six) or lying (no six). ✓ 1

$$P(A|E_1) = \frac{3}{4}, \quad P(A|E_2) = \frac{1}{4}$$

4 Apply Bayes' theorem. ✓ ½

$$P(E_1|A) = \frac{\frac{1}{6} \cdot \frac{3}{4}}{\frac{1}{6} \cdot \frac{3}{4} + \frac{5}{6} \cdot \frac{1}{4}}$$

5 Simplify. ✓ ½

$$P(E_1|A) = \frac{\frac{3}{24}}{\frac{3}{24} + \frac{5}{24}} = \frac{3}{8}$$

6 Required probability $= \frac{3}{8}$.

Answer $P(E_1|A) = \frac{3}{8}$

Watch out: Answering $\frac{3}{4}$: that ignores how unlikely a six is in the first place.

17 **Long answer** Board-pattern 5 marks

Two numbers are selected at random (without replacement) from the first six positive integers. Let X denote the larger of the two numbers obtained. Find the probability distribution of X and hence find its mean $E(X)$.

Formula / idea: Mean $E(X) = \sum x_i p_i$.

1 Number of ways to select 2 numbers from 6. ✓ ½

${}^6C_2 = 15$ equally likely pairs

2 The larger number can be 2, 3, 4, 5 or 6. ✓ ½

$X = 2, 3, 4, 5, 6$

3 Larger number $= x$: the other number is one of $1, \ldots, x-1$. ✓ 1

$$P(X=2) = \frac{1}{15}, \quad P(X=3) = \frac{2}{15}, \quad P(X=4) = \frac{3}{15}$$

4 Continue the same way. ✓ ½

$$P(X=5) = \frac{4}{15}, \quad P(X=6) = \frac{5}{15}$$

5 Write the probability distribution. ✓ ½

X	2	3	4	5	6
$P(X)$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{4}{15}$	$\frac{5}{15}$

6 Check: probabilities add to 1.
$\frac{1+2+3+4+5}{15} = \frac{15}{15} = 1$

7 Write the mean formula and substitute. ✓ 1
$E(X) = \sum x_i p_i = \frac{2(1)+3(2)+4(3)+5(4)+6(5)}{15}$

8 Simplify. ✓ 1
$E(X) = \frac{2+6+12+20+30}{15} = \frac{70}{15} = \frac{14}{3}$

9 Mean of $X = \frac{14}{3}$.

Answer X: $2, 3, 4, 5, 6$ **with** $P(X)$: $\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}$; $E(X) = \frac{14}{3}$

Watch out: Including $X = 1$: the larger of two different numbers is at least 2.

18 Short answer Board-pattern 3 marks

A random variable X has the probability distribution X: $0, 1, 2, 3, 4$ with $P(X)$: $k,\ k,\ 2k^2,\ 3k^2,\ 5k^2 + k$. Find k, $P(X < 2)$ and $P(X \geq 3)$.

Formula / idea: $\sum p_i = 1$, and every $p_i \geq 0$.

1 Sum of all probabilities is 1. ✓ ½
$k + k + 2k^2 + 3k^2 + 5k^2 + k = 1 \ \Rightarrow\ 10k^2 + 3k - 1 = 0$

2 Factorise.
$(5k-1)(2k+1) = 0 \ \Rightarrow\ k = \frac{1}{5}$ or $k = -\frac{1}{2}$

3 Reject $k = -\frac{1}{2}$, since a probability cannot be negative. ✓ ½
$k = \frac{1}{5}$

4 Find $P(X < 2)$. ✓ 1
$P(X < 2) = P(X = 0) + P(X = 1) = 2k = \frac{2}{5}$

5 Find $P(X \geq 3)$. ✓ 1
$P(X \geq 3) = 3k^2 + 5k^2 + k = \frac{8}{25} + \frac{1}{5} = \frac{13}{25}$

6 Hence $k = \frac{1}{5}$, $P(X < 2) = \frac{2}{5}$, $P(X \geq 3) = \frac{13}{25}$.

Answer $k = \frac{1}{5}$, $P(X < 2) = \frac{2}{5}$, $P(X \geq 3) = \frac{13}{25}$

Watch out: Keeping $k = -\frac{1}{2}$: then $P(X = 0)$ would be negative.

19 **Long answer** Board-pattern 5 marks

Assume that the chances of a patient having a heart attack are 40%. A meditation and yoga course reduces the risk of heart attack by 30%, and prescription of a certain drug reduces its chance by 25%. At a time, a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options, a patient selected at random suffers a heart attack. Find the probability that the patient followed a course of meditation and yoga.

Formula / idea: Bayes' theorem: $P(E_1|A) = \dfrac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$.

1	Let E_1: yoga course, E_2: drug, A: patient has a heart attack.	✓ ½
2	Both options are equally likely. $P(E_1) = P(E_2) = \dfrac{1}{2}$	✓ ½
3	Yoga cuts the 40% risk by 30% of itself. $P(A\|E_1) = 0.40 \times (1 - 0.30) = 0.28$	✓ 1
4	The drug cuts the 40% risk by 25% of itself. $P(A\|E_2) = 0.40 \times (1 - 0.25) = 0.30$	✓ 1
5	Apply Bayes' theorem. $P(E_1\|A) = \dfrac{\frac{1}{2} \times 0.28}{\frac{1}{2} \times 0.28 + \frac{1}{2} \times 0.30}$	✓ 1
6	Simplify. $P(E_1\|A) = \dfrac{0.28}{0.58} = \dfrac{28}{58} = \dfrac{14}{29}$	✓ 1
7	Required probability $= \dfrac{14}{29}$.	

Answer $P(E_1|A) = \dfrac{14}{29}$

Watch out: Taking $P(A|E_1) = 0.40 - 0.30 = 0.10$: 'reduces by 30%' means 30% of 0.40.

20 **Case study** CBSE SQP 2025-26 4 marks

Case Study: Excessive use of screens can result in vision problems, obesity, sleep disorders, anxiety and low retention problems. In a class of students of the age group 14 to 17, the students were put into three groups from a feedback form. The first group spent more than 4 hours per day on mobile or gaming screens, the second group spent 2 to 4 hours per day and the third group spent less than 2 hours per day. The first group (high screen time) is 60% of all the students, the second group (moderate screen time) is 30% and the third group (low screen time) is only 10%. It was observed that 80% of the first group, 70% of the second group and 30% of the third group faced severe anxiety and low retention issues.

(i) What is the total percentage of students who suffer from anxiety and low retention issues in the class? 2 marks

(ii) A student is selected at random, and he is found to suffer from anxiety and low retention issues. What is the probability that he/she spends screen time of more than 4 hours per day? 2 marks

Formula / idea: Total probability for (i); Bayes' theorem for (ii).

(i) 2 marks

1	Let E_1, E_2, E_3: student in group 1, 2, 3; A: student has the issues.	
2	Write the prior probabilities. $P(E_1) = 0.6, \quad P(E_2) = 0.3, \quad P(E_3) = 0.1$	✓ ½
3	Write the conditional probabilities. $P(A\|E_1) = 0.8, \quad P(A\|E_2) = 0.7, \quad P(A\|E_3) = 0.3$	✓ ½
4	Theorem of total probability. $P(A) = 0.6 \times 0.8 + 0.3 \times 0.7 + 0.1 \times 0.3 = 0.48 + 0.21 + 0.03$	✓ ½
5	Simplify and convert to a percentage. $P(A) = 0.72 = 72\%$	✓ ½

Answer (i) 72%

(ii) 2 marks

1	Required $P(E_1\|A)$; write Bayes' theorem. $P(E_1\|A) = \dfrac{P(E_1)P(A\|E_1)}{P(A)}$	✓ ½
2	Substitute, using $P(A) = 0.72$ from (i). $P(E_1\|A) = \dfrac{0.6 \times 0.8}{0.72} = \dfrac{0.48}{0.72}$	✓ 1
3	Simplify. $P(E_1\|A) = \dfrac{48}{72} = \dfrac{2}{3}$	✓ ½

4 Required probability $= \frac{2}{3}$.

Answer (ii) $\frac{2}{3}$

Watch out: Answering 0.8 in (ii): that is $P(A|E_1)$, not $P(E_1|A)$.